There are two problems with using tables to design digital filters. First, tables have a limited choice of parameters. For instance, Table 20-1 only provides 12 different cutoff frequencies, a maximum of 6 poles per filter, and *no* choice of passband ripple. Without the ability to select parameters from a continuous range of values, the filter design cannot be *optimized*. Second, the coefficients must be manually transferred from the table into the program. This is very time consuming and will discourage you from trying alternative values.

Instead of using tabulated values, consider including a subroutine in your program that *calculates* the coefficients. Such a program is shown in Table 20-4. The good news is that the program is relatively simple in structure. After the four filter parameters are entered, the program spits out the "a" and "b" coefficients in the arrays A[ ] and B[ ]. The bad news is that the program calls the subroutine in Table 20-5. At first glance this subroutine is really ugly. Don't despair; it isn't as bad as it seems! There is one simple branch in line 1120. Everything else in the subroutine is straightforward number crunching. Six variables enter the routine, five variables leave the routine, and fifteen temporary variables (plus indexes) are used within. Table 20-5 provides two sets of test data for debugging this subroutine. Chapter 31 discusses the operation of this program in detail.

Step Response Overshoot

Butterworth and Chebyshev filters have an overshoot of 5 to 30% in their step responses, becoming larger as the number of poles is increased. Figure 20-3a shows the step response for two example Chebyshev filters. Figure (b) shows something that is unique to digital filters and has no counterpart in analog electronics: the amount of overshoot in the step response depends to a small degree on the cutoff frequency of the filter. The excessive overshoot and ringing in the step response results from the Chebyshev filter being optimized for the *frequency domain* at the expense of the *time domain*.

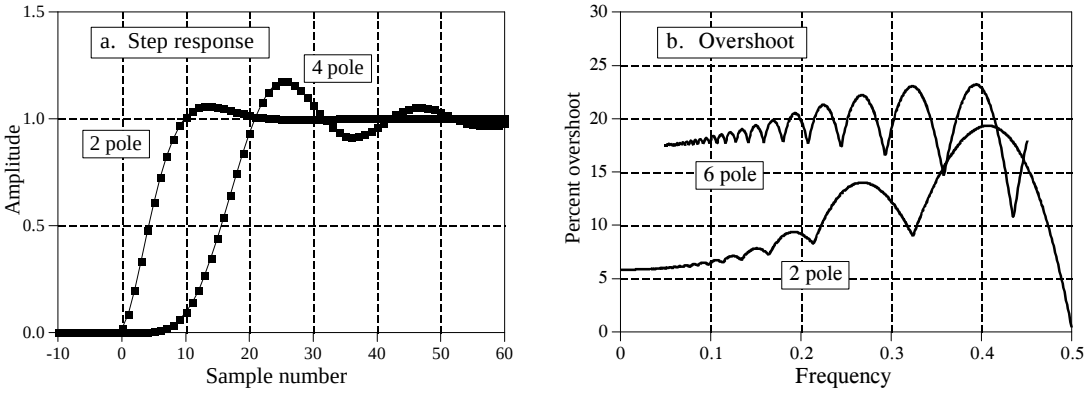

FIGURE 20-3
Chebyshev step response. The overshoot in the Chebyshev filter's step response is 5% to 30%, depending on the number of poles, as shown in (a), and the cutoff frequency, as shown in (b). Figure (a) is for a cutoff frequency of 0.05, and may be scaled to other cutoff frequencies.